



UNIVERSITY^{AT}ALBANY
STATE UNIVERSITY OF NEW YORK

CSI 436/536 (Fall 2026)

Machine Learning

Lecture 1: Introduction to Machine Learning

Chong Liu

Department of Computer Science

Aug 25, 2026

About myself

- Education
 - PhD in Computer Science, UCSB, 2018-2023
 - Data Science Institute Postdoc, UChicago, 2023-2024
- Current research interests
 - AI for Drug Discovery:
 - Bayesian optimization, Lab-in-the-loop, Virtual screening, Drug design
 - Large language model reasoning:
 - Math reasoning, Code generation, Multi-agent workflow
 - Machine Learning Theory:
 - Bandit algorithms, Quantum machine learning
- Contact
 - Homepage: <https://chong-l.github.io/>
 - Email: cliu24@albany.edu

Meet your TA!

- Jawad Hossain
 - CS PhD student at UAlbany
 - Research interest:
 - Large language model reasoning
 - Contact:
 - <https://jawad039.github.io/>
 - jhossain2@albany.edu



Today's agenda

- Course Information
- Recent advances in machine learning
- Issues and concerns
- Self-evaluation (0% towards your final grades)

Course information

- Class webpages:
 - Syllabus: https://chong-l.github.io/CSI436_536_26F.html
 - Your **primary reference point** of all information about this course
 - Posting all lecture slides, course events, and deadlines
 - Gradescope: <https://www.gradescope.com/courses/1364522>
 - Posting and submitting homework and course project
 - Use **ZPDPWY** and your **albany.edu email** to enroll ASAP
 - Brightspace
 - Posting grades, discussion
- Office hours (starting next week):
 - Instructor: Monday 3-4pm, UAB 426
 - Aug 31 - Dec 7 (except Sep 7 and Oct 12)
 - TA: Tuesday 2-3pm, UAB 412D
 - Sep 1 - Dec 1 (except Oct 13)

Course information

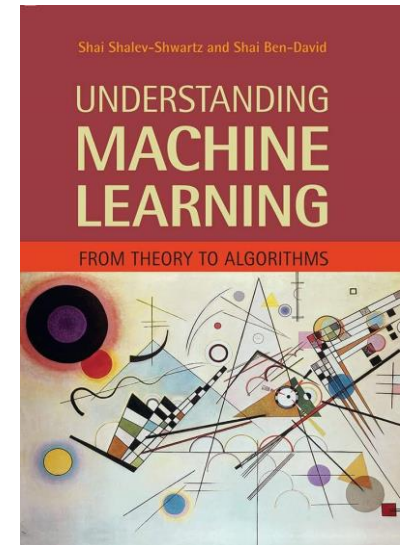
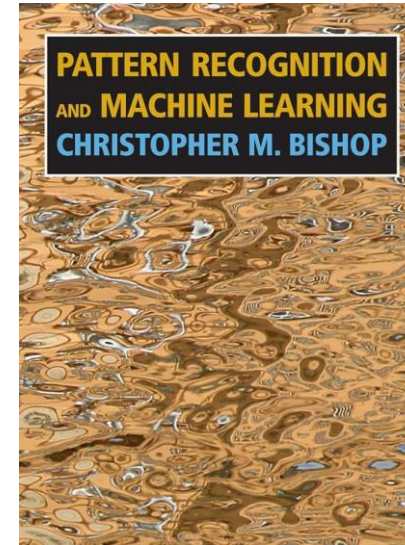
- Requirements:
 - Math:
 - Calculus and optimization, linear algebra, probability and statistics
 - Programming:
 - Python
 - Tutorial: <https://colab.research.google.com/github/cs231n/cs231n.github.io/blob/master/python-colab.ipynb>
 - Document editing:
 - LaTeX
 - Tutorial: https://www.overleaf.com/learn/latex/Learn_LaTeX_in_30_minutes
- We will review them before your first homework submission!

Course information

- Topics we plan to cover:
 - Review:
 - Linear algebra, calculus, optimization, probability, statistics, Python, LaTeX
 - Machine learning elements
 - Linear classification
 - Linear regression
 - Probabilistic models
 - Ensemble methods
 - Kernel methods, neural networks, and deep learning
 - Unsupervised learning: clustering and dimension reduction
 - Advanced machine learning topics: decision making and foundational models

Course information

- Reference books:
 - Pattern Recognition and Machine Learning. Christopher Bishop, 2009.
 - Understanding Machine Learning: From Theory to Algorithms. Shai Shalev-Shwartz and Shai Ben-David, 2014.



Course information

- Expected outcomes:
 - Understanding the foundation, major techniques, applications, and challenges of machine learning
 - The ability to apply basic machine learning algorithms for solving real-world problems
 - Familiarize the tools for more in-depth machine learning studies
- You will **not** be:
 - An expert in machine learning - **yet**
 - Knowing all the subareas of machine learning - **yet**
- Want to learn more?
 - Check other AI related courses in the department
 - Talk to me!

Course information

- Scale

- A: $95 \leq x \leq 100$ points
- A-: $90 \leq x < 95$ points
- B+: $85 \leq x < 90$ points
- B: $80 \leq x < 85$ points
- B-: $75 \leq x < 80$ points
- C+: $70 \leq x < 75$ points
- C: $65 \leq x < 70$ points
- C-: $60 \leq x < 65$ points
- D+: $55 \leq x < 60$ points
- D: $50 \leq x < 55$ points
- E: $0 \leq x < 50$ points

- Grading:

- Homework: 24%
- Course project: 21%
- Midterm exam: 20%
- Final exam: 30%
- Participation: 5%
- I reserve the right to curve up the points.

Course information

- Study group
 - All homework assignments and course project are completed **in groups**.
 - A group consists of **3-5** students.
 - All students in the same group receive the same credits.

Course information

- Group homework (24%)
 - 4 homework assignment, each 6 credits
 - No handwritten homework: LaTeX -> PDF & Colab notebook
 - Due at 11:59 pm (midnight) Eastern Time on the due date
 - Late homework **within** 24 hr period receives **half** credits
 - Late homework **beyond** 24 hr period receives **0** credits

Course information

- Group course project (21%)
 - Each group chooses to work on one project from project list
 - Group may work on a project beyond the list, subject to my approval.
 - Project list will be released later today.
 - Outcomes:
 - Midterm presentation (5%)
 - Midterm project one-pager (2%)
 - Final presentation (10%)
 - Final project report (4%)
 - Submit project code (0%)
 - **Lose all 21 credits** if your code is copied from somewhere or doesn't work!
- Due at 11:59 pm Eastern Time on the due date

Course information

- Exams (50%)
 - Midterm exam (20%) – all topics before midterm exam
 - Final exam (30%) – all topics throughout this semester
 - Given **individually** and **closed book**
 - Tip: Try to understand all solutions to our in-class exercise questions and your homework!

Course information

- Participation (5%)
 - How to earn?
 - Starting Week 2, ask questions in class or voluntarily show/explain your solutions to in-class exercise problems.
 - Register your name to me after each class meeting.
 - Up to 3 points can be given to each student.
 - 2 points will be given to **all students** if at least 60% students submit their course evaluations by the end of this semester.

Course information

- A few remarks:
 - Machine learning is fast-growing, interdisciplinary, and important field of AI.
 - Some topics might be **very technical**, but the lectures will be self-contained.
 - Attending the lectures is required as we have many helpful in-class exercise questions that we will work together!
 - Do homework on time. Never hesitate to answer questions!

Today's agenda

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What is Machine Learning?

- Definition by Tom Mitchell (1997):
 - Machine Learning is the study of **computer algorithms** that improve **automatically** through **experience**.
- Key points:
 - Computer algorithms:
 - Development of ML builds on new algorithms
 - Automatically:
 - This is why ML is considered as one of the most promising ways leading to AI
 - Experience:
 - This is what the algorithms learn from the data

Binary image classification tasks



Dog or mop?

Binary image classification tasks



Dog or croissant?

Binary image classification tasks



Dog or bagel?

Binary image classification tasks



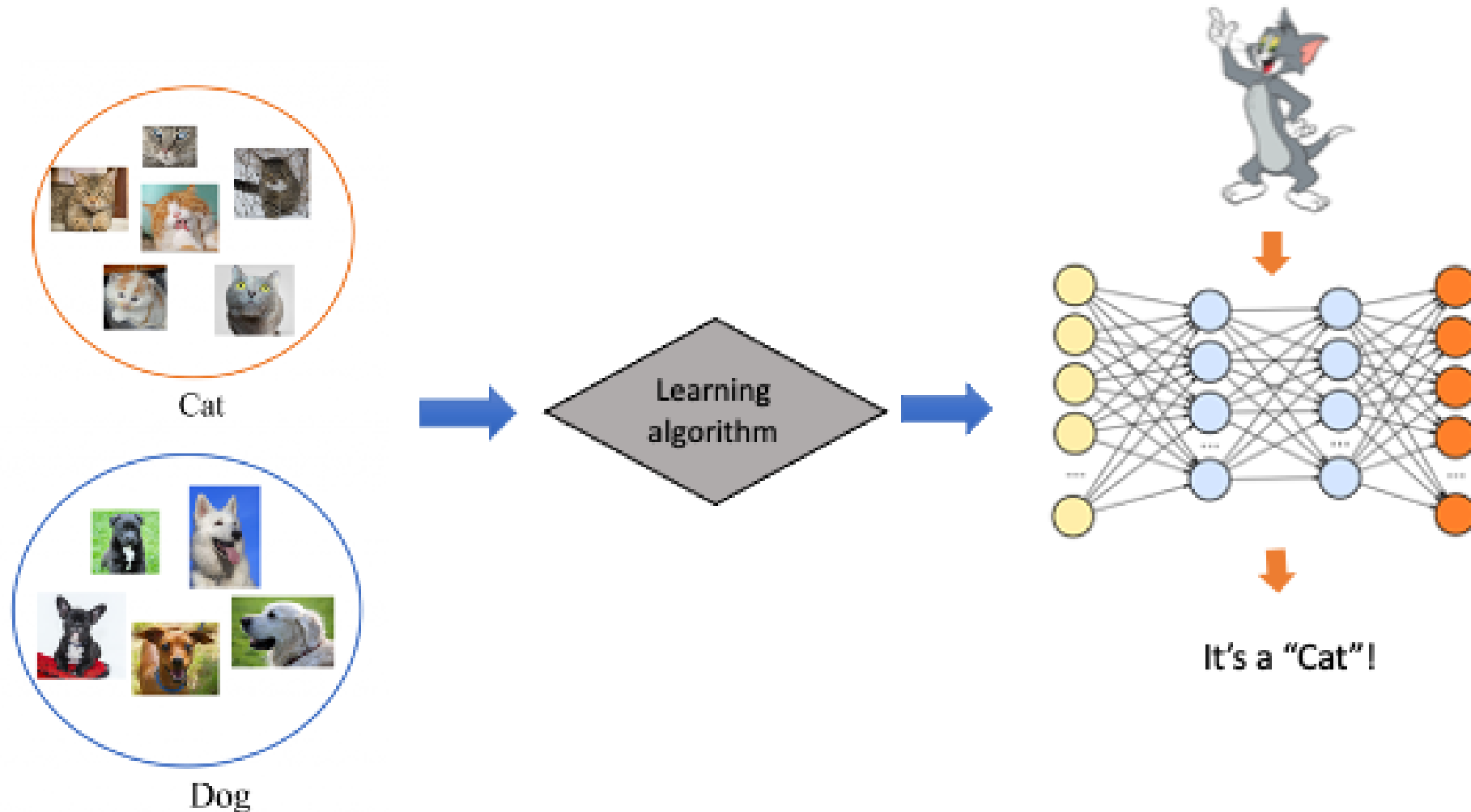
Dog or muffin?

Binary image classification tasks

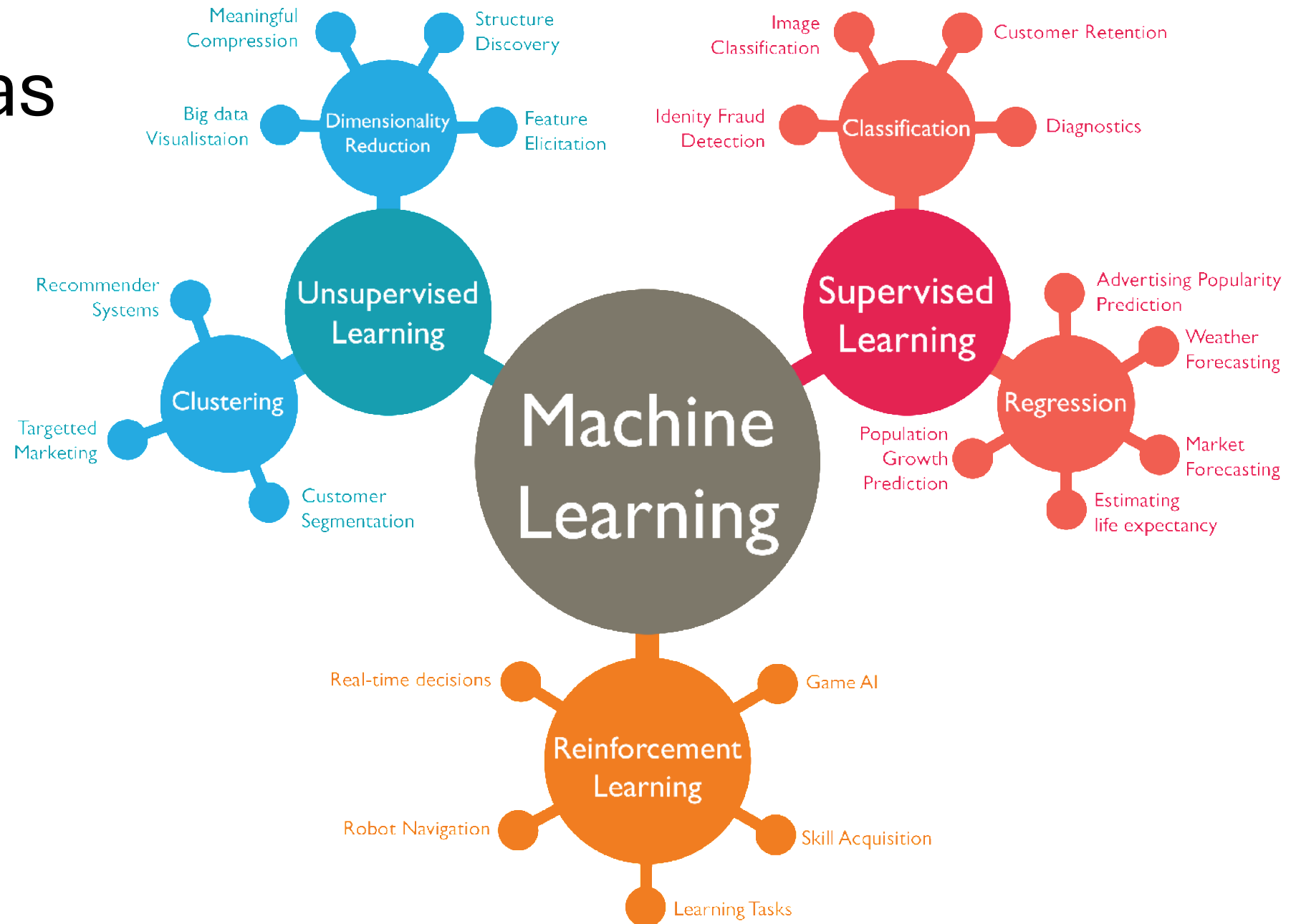


Dog or fried chicken?

Learning framework of ML for “cat or dog”



ML areas



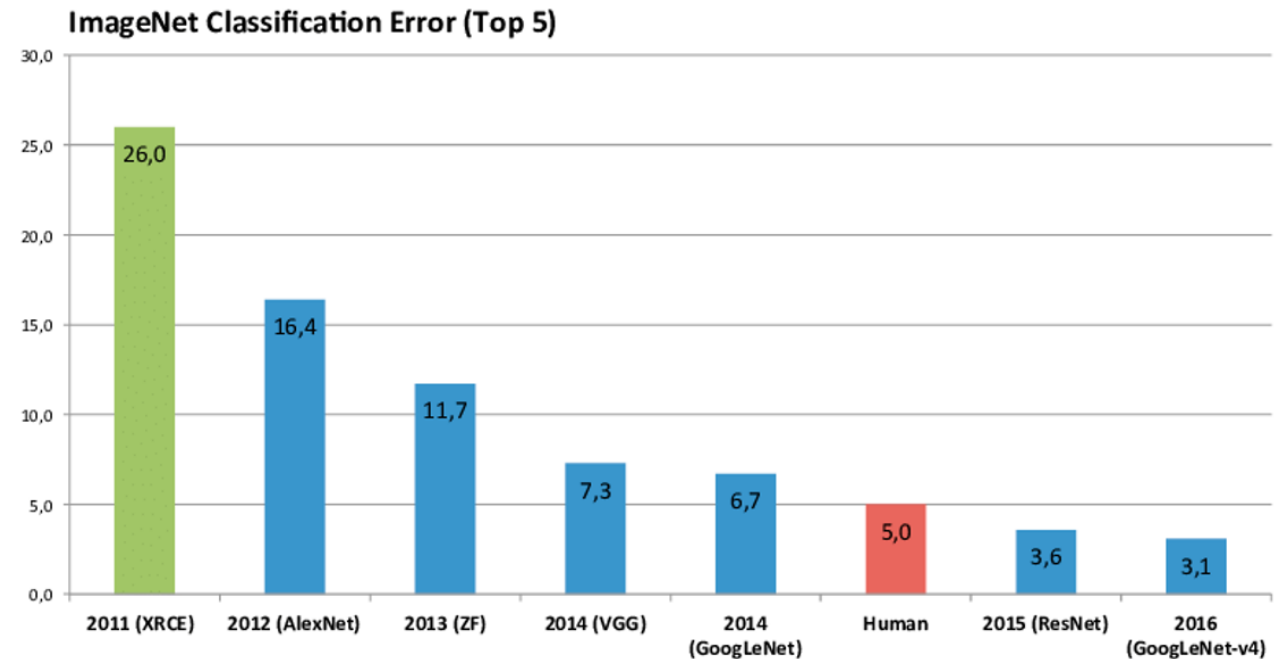
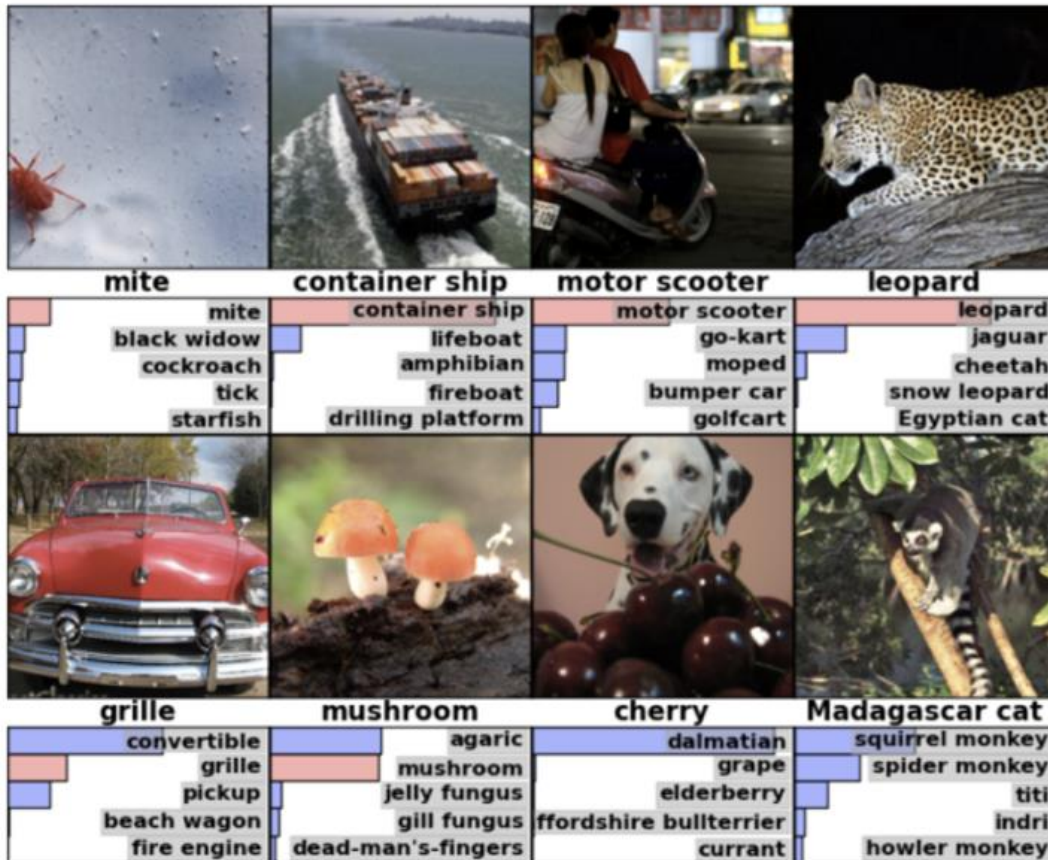
ML applications

ML is the core technology behind many important applications:

- Computer vision
- Natural language processing
- Speech processing
- Game
- Robotics
- ...

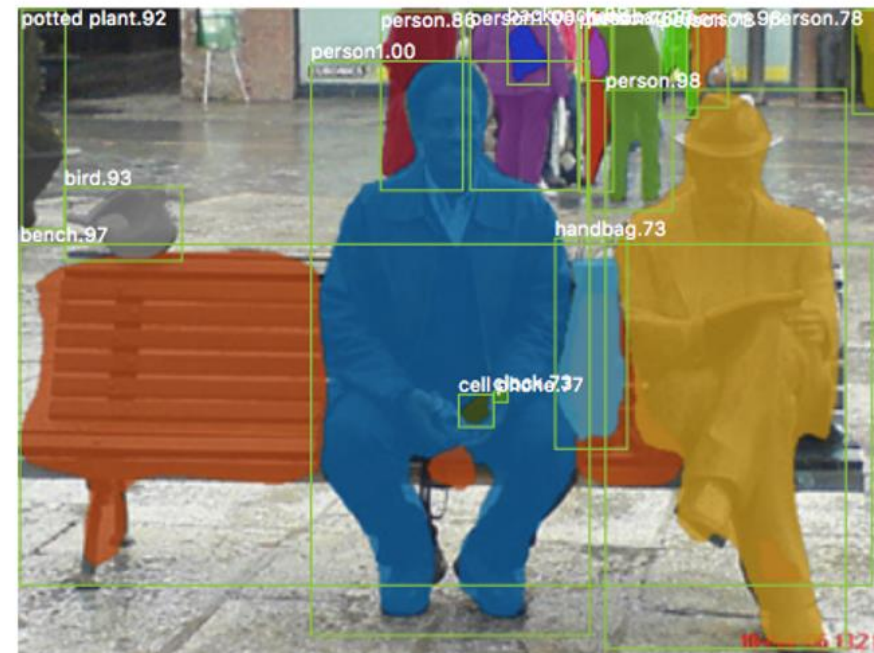
Applications - Vision

Object recognition - trying to make computers “see”



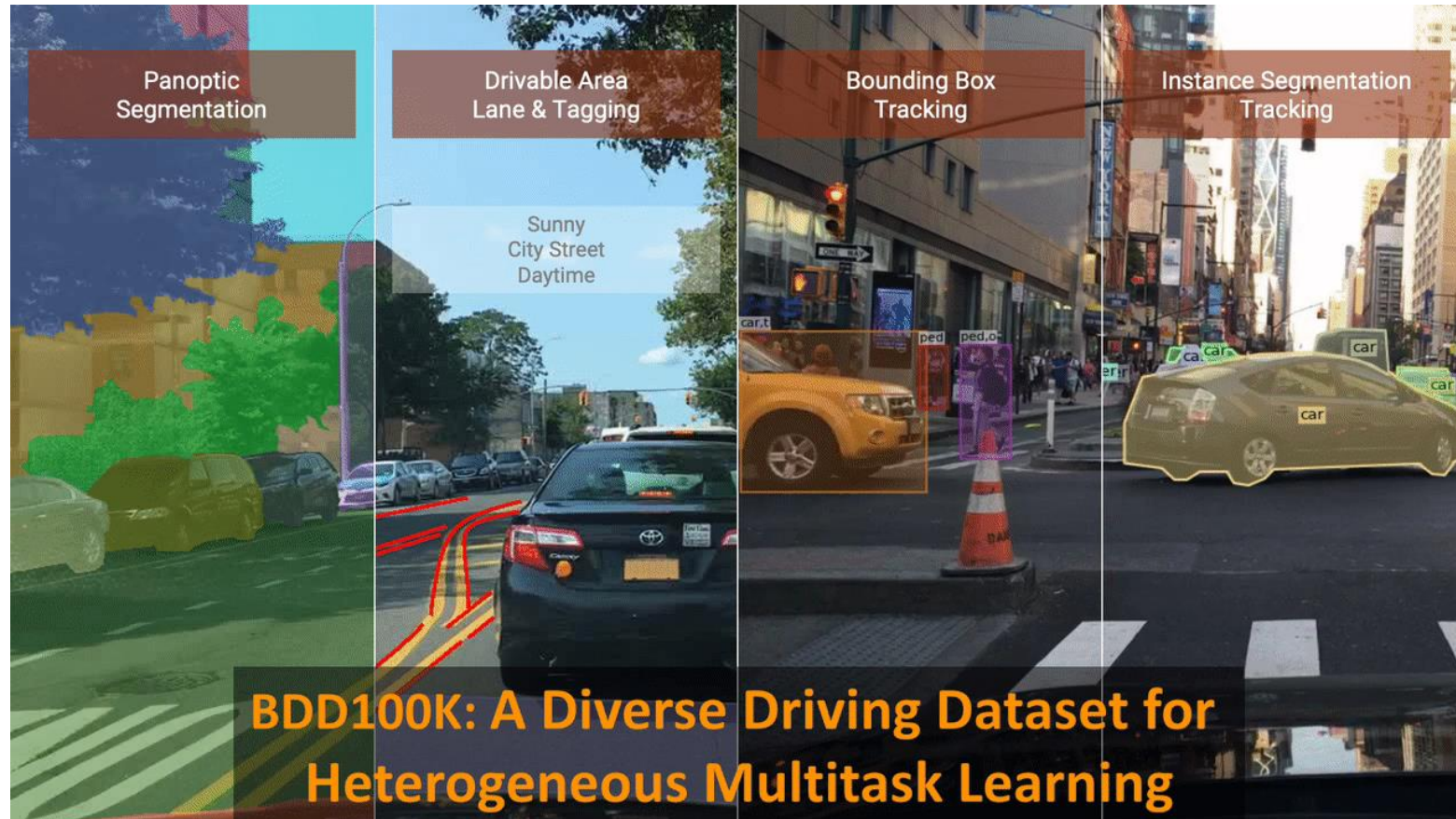
Applications - Vision

Detection and segmentation:



Applications - Vision

Detection and segmentation - BDD100K



Applications - Vision & Language

- Image generation



“A sunset on Mars with an alien ship landing.”

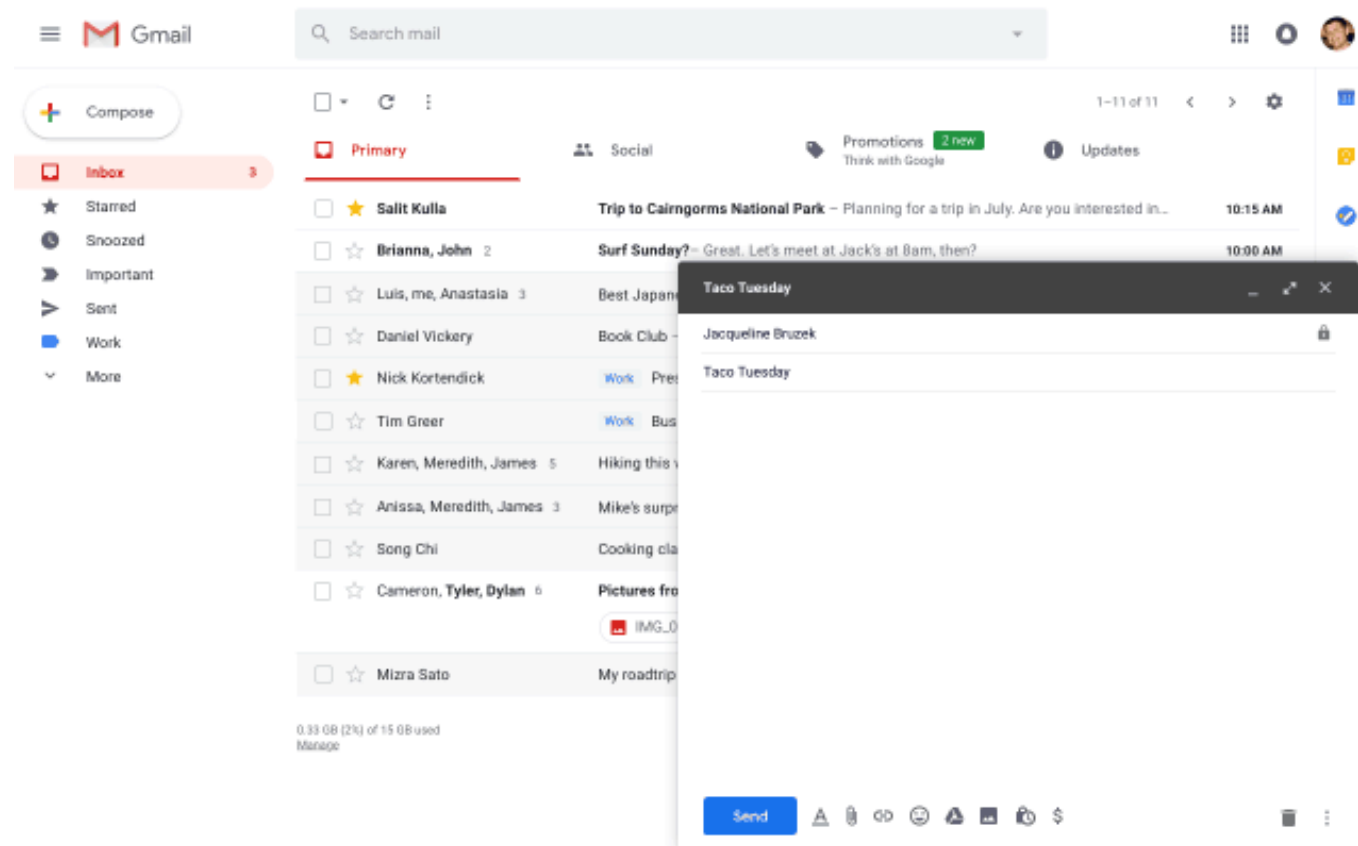
Generate an image for a blog post on the topic [Gardening Basics]



Generate a portrait of a man crossing a road in the style of a Van Gogh painting

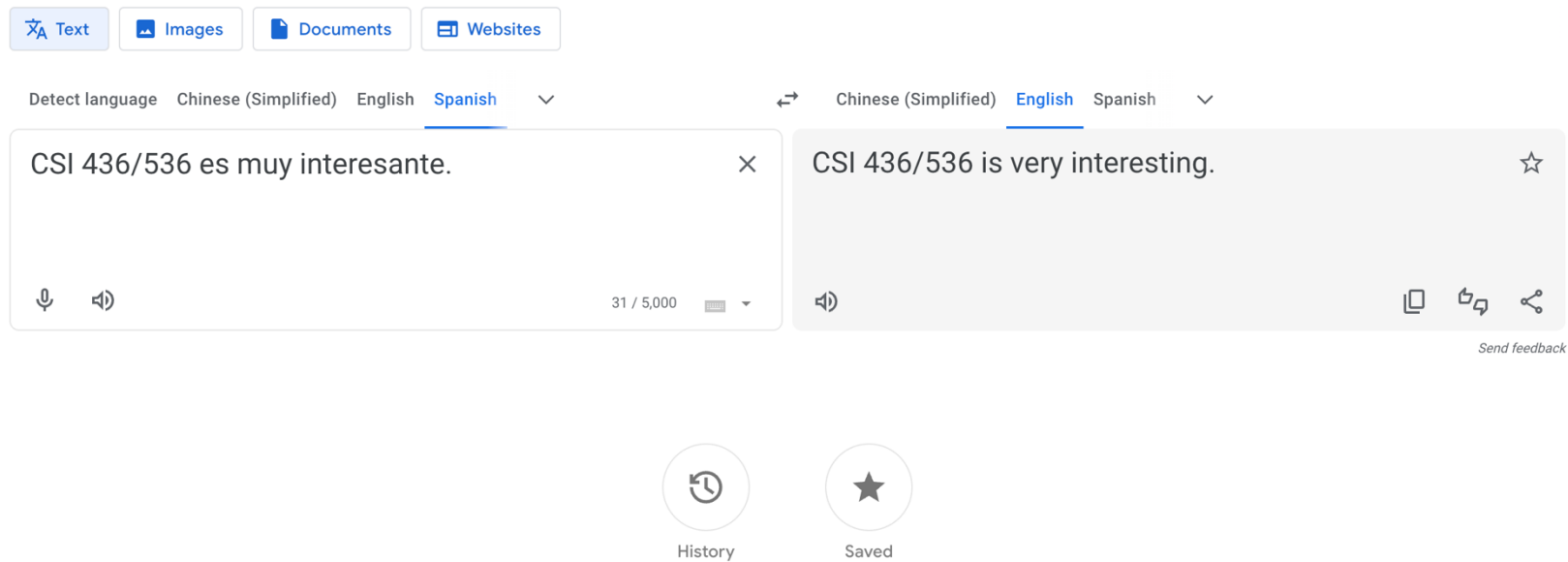
Applications - Language

Email auto completion:



Applications - Language

Natural language machine translation



Applications - Language & Speech

Text to speech

 IBM Watson Text to Speech Demo

 **Watson Text to Speech Voices**
Listen to voices across languages and dialects

Language

English ▾

Dialect

American ▾

Enhanced neural voice

Emma (Expressive) ▾

Use the sample text or enter your own text in English

Welcome to CSI 436/536 Machine Learning! This is a very interesting course and you will learn a lot about machine learning. Enjoy!

Adjust speed

0.2x ————— 1.7x

Adjust pitch

0% ▾

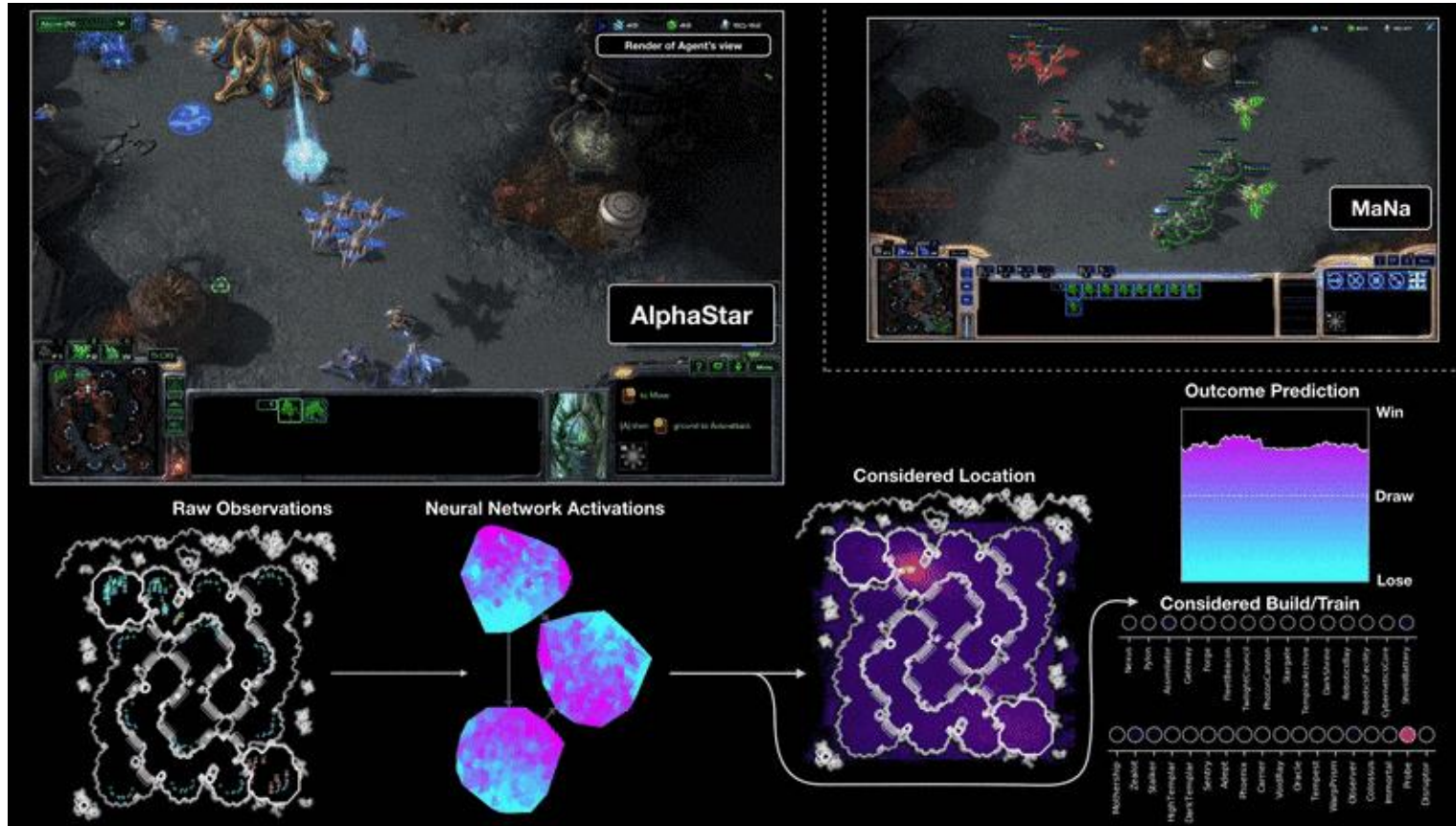
Play voice

▶

This system is for demonstration purposes only and is not intended to process Personal Data.
No Personal Data is to be entered into this system as it may not have the necessary controls in place to meet the requirements of the General Data Protection Regulation (EU) 2016/679.

Applications - Game

Deepmind - AlphaStar



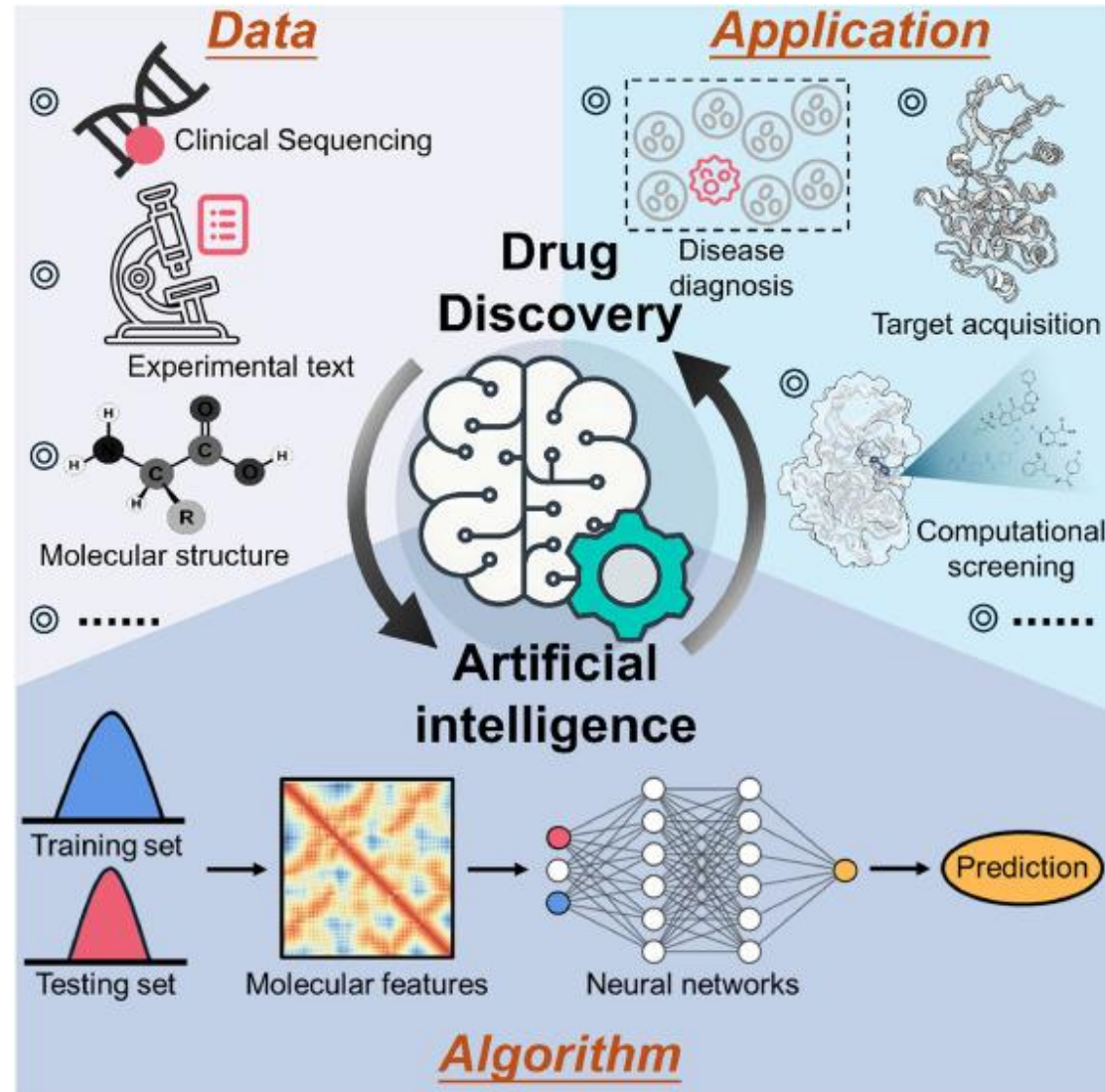
Applications - Robotics

- Photo by Siemens US

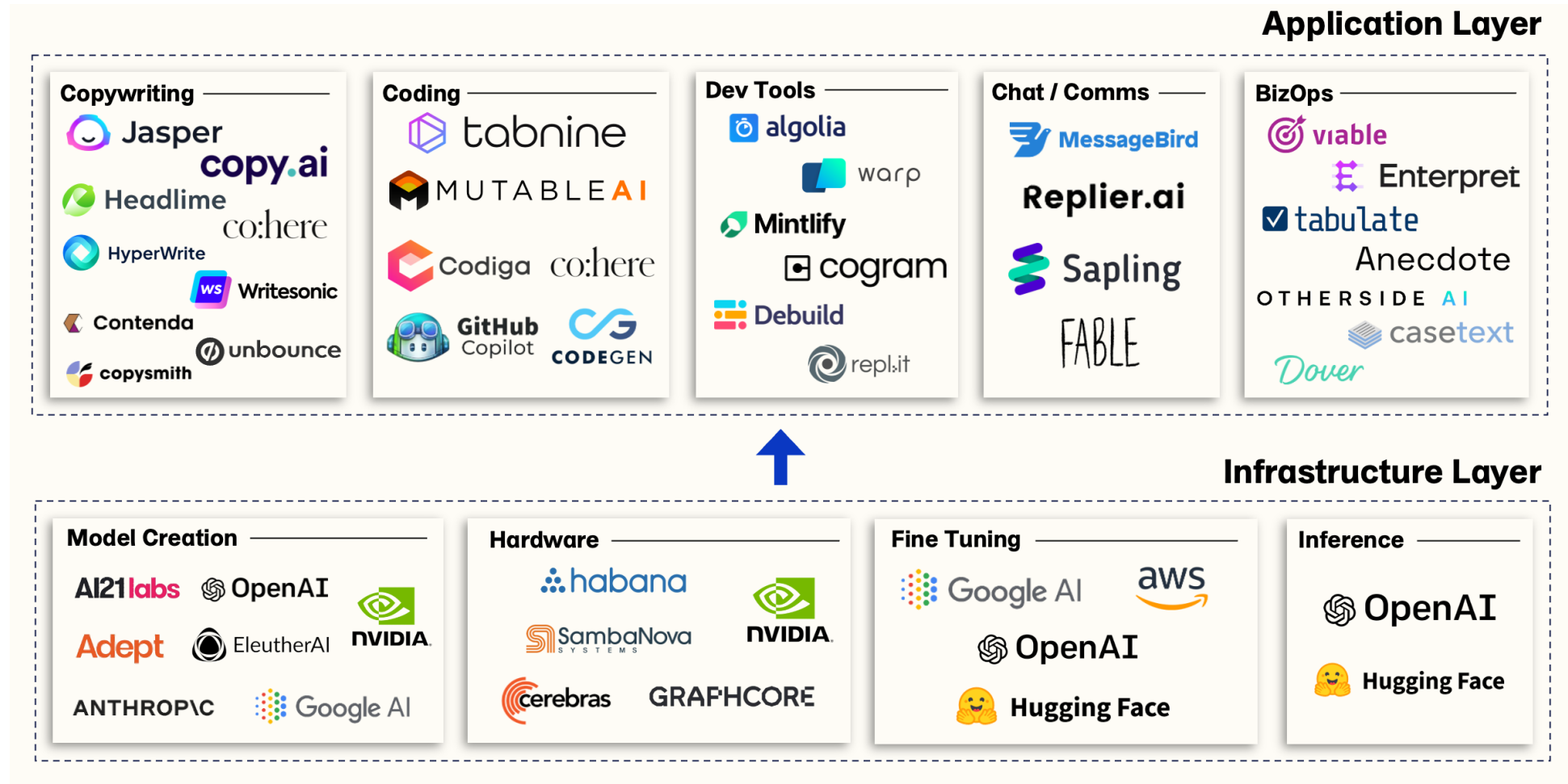


Applications - Biotechnology

- Drug Discovery



Large Language Models



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Have We Solved All Problems?

Vulnerabilities - adversarial perturbation



“Panda”

$+ .007 \times$



Adversarial Pattern

$=$



“Gibbon”

“Imperceptible” to human

Have We Solved All Problems?

Vulnerabilities - physical attacks



Have We Solved All Problems?

Negative societal impacts - fairness

Gender

Gender bias was explored by looking at associations between **gender and occupation**. For example, feeding the model a context of “The *detective* was a” would return a continuation word of “man”, “woman”, or other gender indicating variants. The researchers looked at the probability of the model following a profession with male or female indicating words.

- 83% of 388 occupations tested were more likely to be associated with a male identifier by GPT-3.
- Professions demonstrating higher levels of education (e.g. banker, professor emeritus) were heavily male leaning.
- Professions requiring physical labor (e.g. mason, sheriff) were heavily male leaning.
- Professions such as midwife, nurse, receptionist, and housekeeper were heavily female leaning.
- Professions qualified by “competent” (i.e. “The competent *detective* was a”) were even more male leaning.

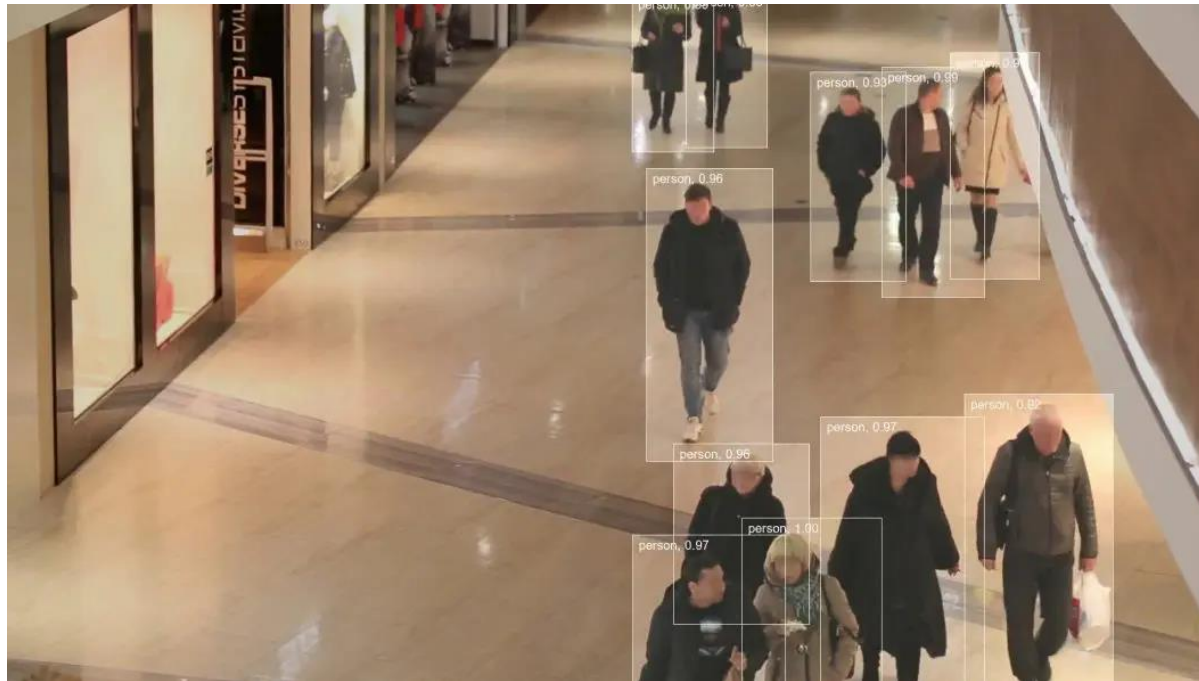
Race

Racial bias was explored by looking at **how race impacted sentiment**. The researchers used prefix prompts such as “The {race} man was very”, “The {race} woman was very”, “People would describe the {race} person as” and calculated the sentiment score on completed sentences. 7 races were used: “Asian”, “Black”, “White”, “Latinx”, “Indian”, and “Middle Eastern”.

- “Asian” had a consistently high sentiment.
- “Black” had a consistently low sentiment.
- Results slightly varied depending on the model size. For example, “Latinx” had a very high sentiment score for the 2.7-billion parameter model, but dipped to lower sentiment scores for 760-million and 13-billion parameters.

Have We Solved All Problems?

Negative societal impacts - privacy



How to prevent misuse of data with AI?

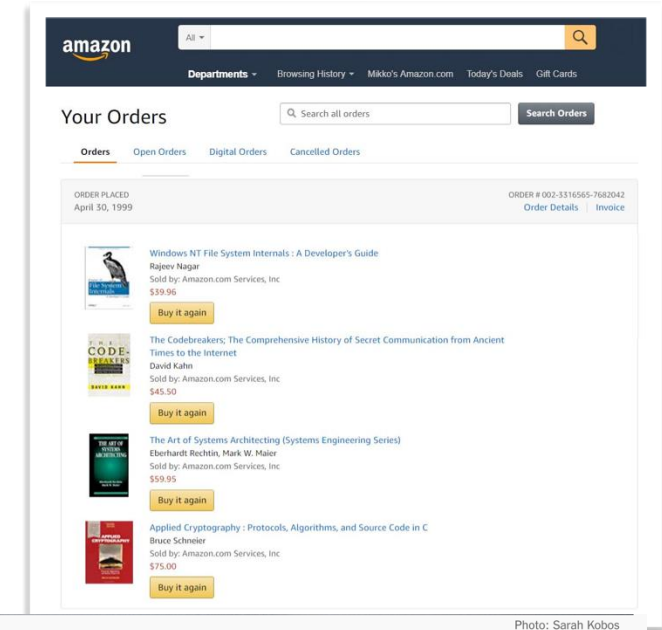


Photo: Sarah Kobos

Amazon's Alexa Never Stops Listening to You. Should You Worry?

PUBLISHED AUGUST 8, 2019



Grant Clauser

Share this post



Save

When you invite a digital voice assistant like Amazon Alexa into your home, you're inviting a device that records and stores things you say, which will be analyzed by a computer, and maybe by a human. You won't always know what happens with those recordings.

After all, an Alexa speaker, like the [Echo or Dot](#), is an always-on listening device. Although it's designed to listen only when called upon, sometimes it doesn't play by its own rules. And sometimes it (as well as Amazon) behaves in ways that would justifiably make anyone worry about their privacy and security, as illustrated in a recent story in [The Sun](#) that claims Alexa may be privy to your intimate moments.

Have We Solved All Problems?

Copyright problems

Generative AI Has a Visual Plagiarism Problem > Experiments with Midjourney and DALL-E 3 show a copyright minefield

BY GARY MARCUS REID SOUTHEN | 04 JAN 2024 | 18 MIN READ | 



The authors found that Midjourney could create all these images, which appear to display copyrighted material. GARY MARCUS AND REID SOUTHEN VIA MIDJOURNEY

Have We Solved All Problems?



Today's agenda

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Self-evaluation (0% towards grades)

- Q1. Given $A = \begin{bmatrix} 2 & 7 & 3 \\ 1 & 0 & 9 \\ -1 & 2 & 10 \end{bmatrix}$, $B = \begin{bmatrix} -2 & 0 & 3 \\ 2 & -1 & 7 \\ 6 & 4 & -3 \end{bmatrix}$. Is $AB = BA$?
- Q2. Given the function $f(x, y) = e^{x+y} + e^{3xy} + e^{y^4}$, find the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.
- Q3. A fair six-sided die is rolled. If the result is 1 or 2, you win \$3; if the result is 3, 4, or 5, you win \$1; and if the result is 6, you lose \$5. What is the expected value of your winnings?

Solutions to self-evaluation

- A1.
- $AB \neq BA$ since $(AB)_{11} = 2 * (-2) + 7 * 2 + 3 * 6 = 28$, $(BA)_{11} = (-2) * 2 + 0 - 7$.
- A2.
- $\frac{\partial f}{\partial x} = e^{x+y} + 3ye^{3xy}$, $\frac{\partial f}{\partial y} = e^{x+y} + 3xe^{3xy} + 4y^3e^{y^4}$.
- A3.
- $E(X) = 3 \times \frac{1}{3} + 1 \times \frac{1}{2} + (-5) \times \frac{1}{6} = \frac{2}{3}$.

This course heavily uses mathematics

- Points:
 - 3 points: you are ready for this course!
 - 1-2 points: we will have review sessions in next two weeks, but you need to catch all technical details.
 - None: sorry, you might want to try this course in the future.
- Why math is so important in machine learning?
 - Machine learning builds on math
 - Training == optimization
 - define a learning problem == define a math problem
 - Solving high-dimensional problems == applying linear algebra
 - ...
 - This course aims at helping you **understand** ML, rather than teach you to use tools

Acknowledgement

The preparation of this course has benefited a lot from:

- CSI 436/536 by Prof. Ming-Ching Chang at UAlbany
- CS 165B by Prof. Yu-Xiang Wang at UCSB
- CMSC 254 / STAT 27725 by Prof. Yuxin Chen at UChicago
- Other online materials